NovaNote - Voice of the Customer

Exit interview analysis | portfolio project A1

1. Executive summary

NovaNote, a Notion-class productivity suite, is losing business accounts. This study analyzes 10 exit interviews (50 coded segments, 13 themes). Overall exit sentiment is mildly negative (mean -0.08) with 50% of segments negative. Churn concentrates in support, collaboration, roadmap and integrations; onboarding and the core editor are praised. Founders exit more negative (-0.06) than product managers (+0.02). The theme network splits into three clusters: API/integrations, cost-and-trust, and scale/collaboration/support. Recommendations: a support SLA with a public bug roadmap, document-level permissions, stable APIs and transparent pricing.

metric	value
segments	50
interviews	10
themes	13
mean sentiment	-0.08
segments negative	50%
founders mean	-0.06
PMs mean	0.02

2. Business problem

NovaNote (a Notion competitor) is seeing above-expected churn among small and mid-size business accounts. Leadership needs evidence on which product and service dimensions drive cancellations, which segments are most negative, and what a retention roadmap should prioritize first.

3. Data analysis strategy

An interpretive-to-quantitative pipeline was applied: (1) convert interviews into a coded segmented corpus; (2) measure sentiment robustly with a pretrained transformer cross-checked against a lexicon model and human gold labels; (3) model theme relationships as a weighted co-occurrence network with centrality, community detection and propagation indicators; (4) surface everything in an interactive dashboard and a written report.

4. Technological stack (all free / open-source)

tool	role
Python 3.12	processing
pandas / numpy	data handling
transformers + RoBERTa	sentiment model
VADER (NLTK)	cross-check
networkx + python-louvain	graph, centrality, communities
scipy / scikit-learn	metrics / stats
matplotlib / wordcloud	static figures
Streamlit + Plotly	interactive dashboard
reportlab	PDF report
GitHub	hosting & versioning

5. Methodology

Corpus: 10 fictional-but-realistic exit interviews (5 founders, 5 product managers), split into 50 paragraph segments. Each segment was coded multi-label against a 13-theme dictionary using regex keyword matching.

Sentiment: per segment, cardiffnlp/twitter-roberta-base-sentiment (CPU) returns a positive/neutral/negative distribution scored [-1, +1]; VADER supplies a lexicon baseline. Reliability is reported against a hand-labeled gold set (accuracy, macro-F1, Cohen's kappa) and model agreement is audited via Cohen's kappa.

Network: a weighted co-occurrence graph connects themes that co-occur in the same segment. Centrality (degree, weighted degree, betweenness, closeness, eigenvector), PageRank as an influence proxy, also strong-tie reach. Communities were found with Louvain. Layout is a reproducible 3D force-directed embedding.

6. Results

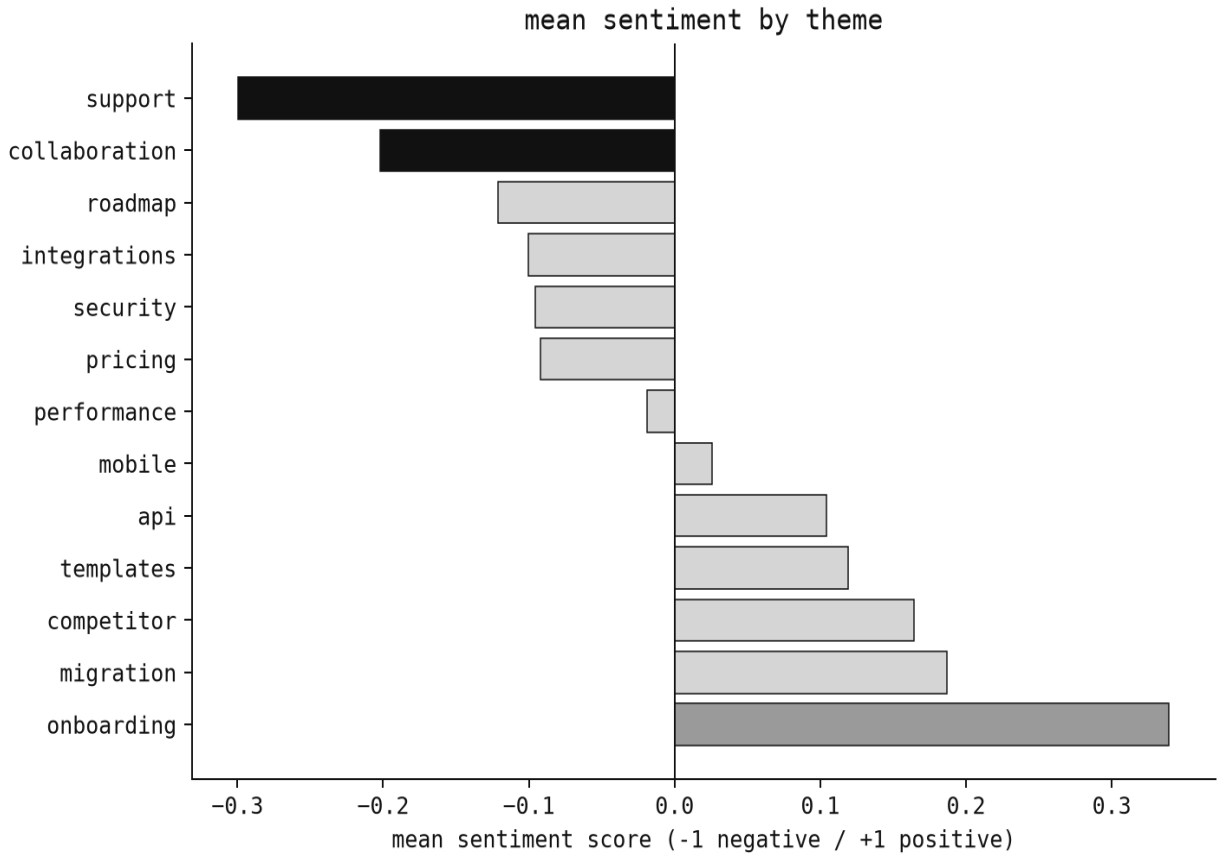
6.1 Sentiment model reliability

model	accuracy	macro_f1	cohen_kappa
roberta	0.6	0.469	0.356
vader	0.4	0.287	0.153
roberta_vader	nan	nan	0.674

RoBERTa outperforms the lexicon baseline on every metric and is retained as the primary engine. The confusion matrix against gold labels:

pred:negative	pred:neutral	pred:positive
24	4	2
1	3	12
0	1	3

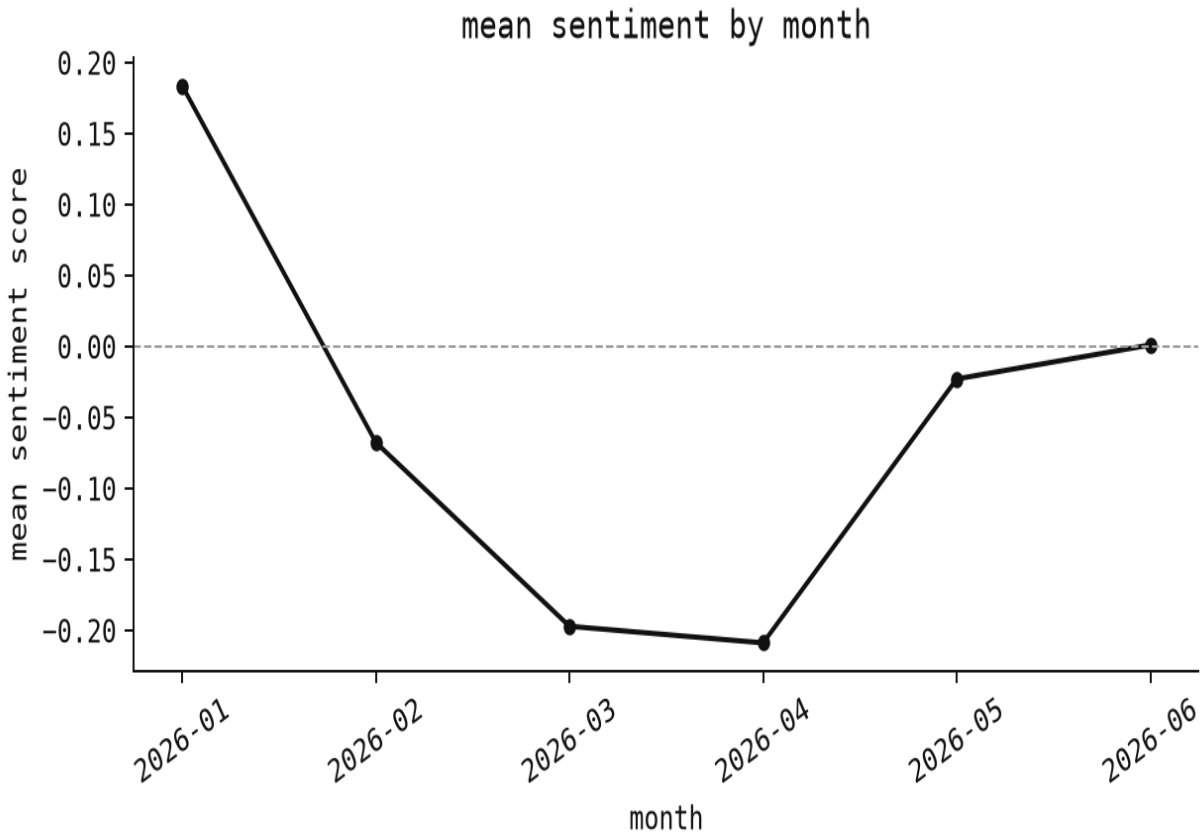
6.2 Sentiment by theme

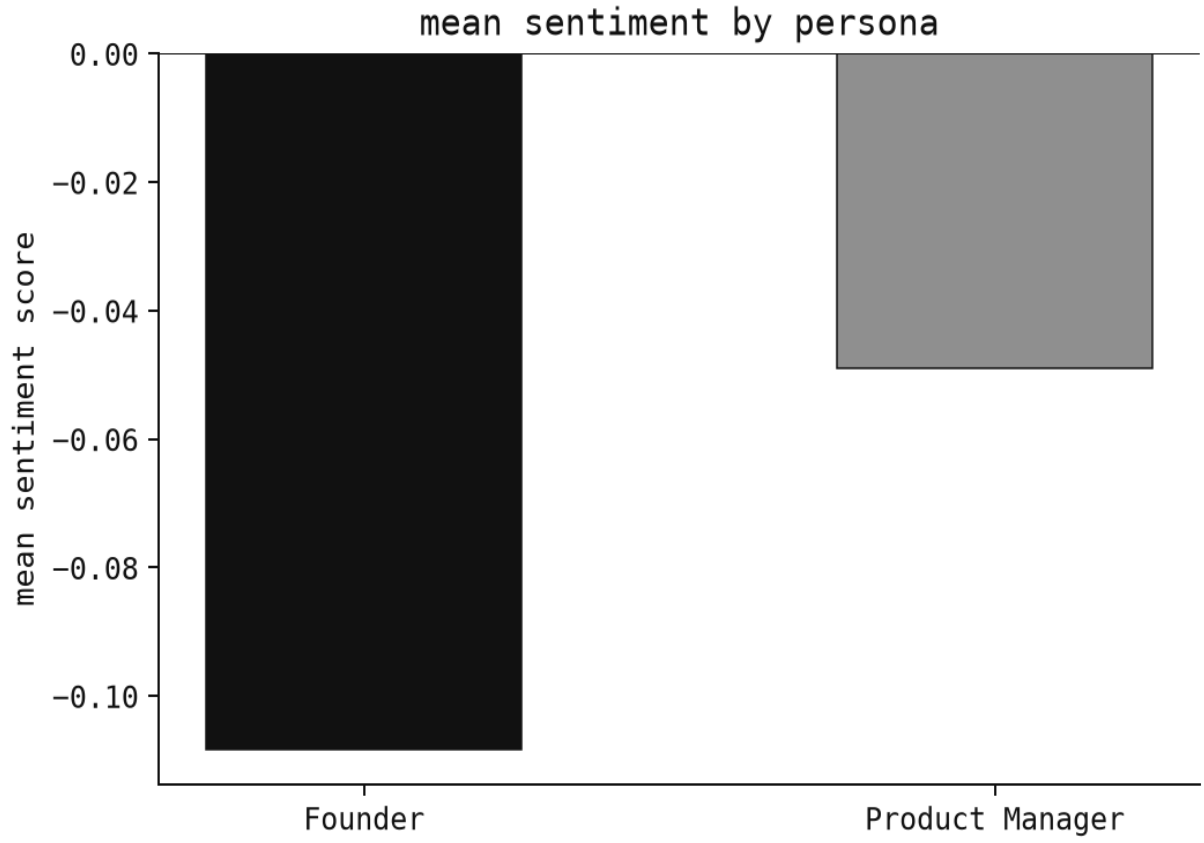


theme	mean_score	n_segments	negative_share
support	-0.3	15	0.6
collaboration	-0.2	9	0.56
roadmap	-0.12	9	0.56
integrations	-0.1	14	0.5
security	-0.1	4	0.25
pricing	-0.09	18	0.5
performance	-0.02	18	0.44
mobile	0.03	7	0.43
api	0.1	6	0.5
templates	0.12	7	0.43
competitor	0.16	6	0.33
migration	0.19	16	0.31
onboarding	0.34	7	0.29

Most negative: support, collaboration, roadmap. Most positive: migration, onboarding.

6.3 Sentiment trends

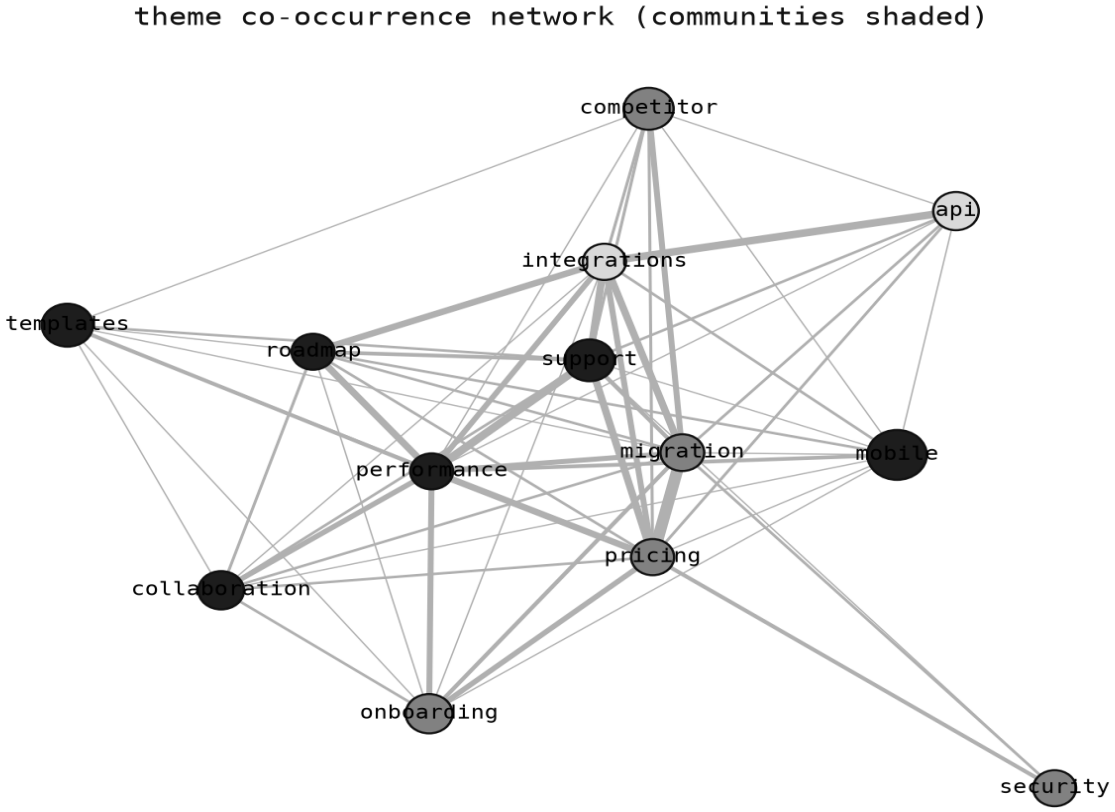




month	score
2026-01	0.1834531188011169

month	score
2026-02	-0.0678593754768371
2026-03	-0.1973242282867431
2026-04	-0.2090366885066032
2026-05	-0.0230613946914672
2026-06	0.0008970126509666

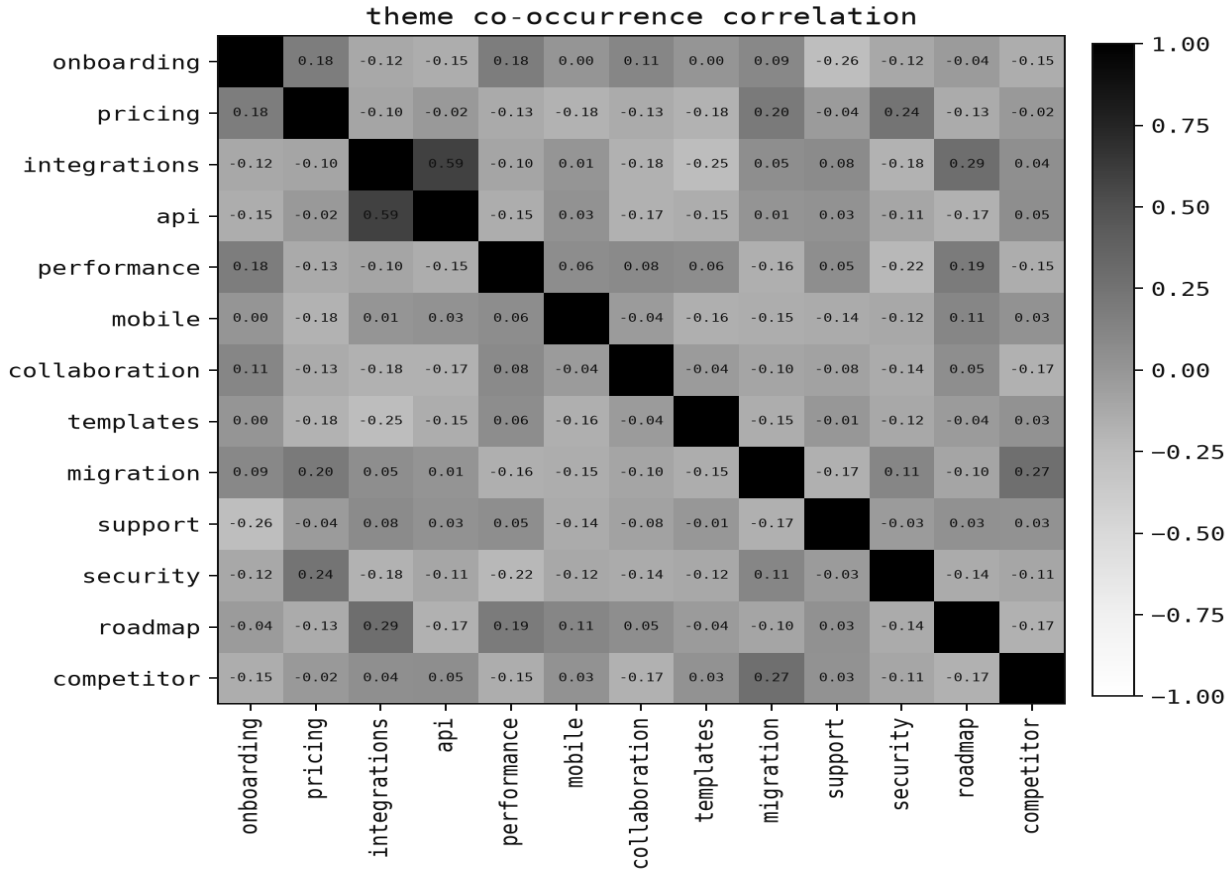
6.4 Theme network, centrality and communities



theme	degree	weighted_degree	betweenness	closeness	eigenvector	pagerank	community
mobile	10	14	0.344	0.857	0.157	0.051	2
templates	8	11	0.158	0.75	0.126	0.042	2
competitor	8	14	0.132	0.75	0.167	0.051	1
support	11	32	0.131	0.923	0.352	0.103	2
onboarding	8	17	0.071	0.75	0.204	0.059	1
collaboration	9	17	0.05	0.8	0.196	0.059	2
api	7	15	0.046	0.706	0.187	0.053	0
migration	12	37	0.016	1.0	0.385	0.118	1
integrations	10	35	0.004	0.857	0.372	0.11	0
pricing	12	39	0.003	1.0	0.404	0.125	1
performance	11	40	0.0	0.923	0.404	0.126	2
security	3	6	0.0	0.571	0.083	0.028	1
roadmap	9	23	0.0	0.8	0.269	0.076	2

Communities (Louvain):

community	themes
0	integrations, api
1	onboarding, pricing, migration, security, competitor
2	performance, mobile, collaboration, templates, support, roadmap



7. Recommendations

CLUSTER 0 [integrations, api]: API & integrations cluster - stabilize the public API (SDK, rate limits), ship native connectors and publish an integration roadmap.

CLUSTER 1 [onboarding, pricing, migration, security, competitor]: Cost & trust cluster - transparent pricing with no surprise billing, clearer plan feature tables, official import/export, and published security/residency + SSO docs.

CLUSTER 2 [performance, mobile, collaboration, templates, support, roadmap]: Scale & collaboration cluster - support reply SLA and public known-bug status, document-level permissions, assignable comments, versionable locked templates, offline-first mobile, and performance work at enterprise scale.

Priority sequence: (1) support SLA + public roadmap, (2) pricing transparency, (3) performance at scale, (4) collaboration permissions, (5) API stabilization.